数学分析(三)第五、六周作业

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注: 题目按照教材《数学分析(第三册)》配套的全部习题的顺序排列,题号右上角有星号的题目表示出现在布置的作业题中.

14.25*. 称 $f(\boldsymbol{x})$ 是 K **次齐次函数** $(K \in \mathbb{R})$, 如果 $f(\boldsymbol{x})$ 定义在区域 $D \subseteq \mathbb{R}^n$ 上, 且 $\forall t > 0, \forall \boldsymbol{x} \in D$, 都有 $f(t\boldsymbol{x}) = t^K f(\boldsymbol{x})$. 设 K 次齐次函数 $f(\boldsymbol{x})$ 在 D 内具有各个 $k \in \mathbb{Z}^+$ 阶连续偏导数, 求证:

$$\left(\sum_{i=1}^{n} x_i \frac{\partial}{\partial x_i}\right)^k f(\boldsymbol{x}) = K(K-1)\cdots(K-k+1) f(\boldsymbol{x}).$$

解答:

断言: $\forall k \in \mathbb{Z}^+$, 都有

$$\frac{\partial^k f(t\boldsymbol{x})}{\partial t^k} = \left(\sum_{i=1}^{n} x_i \frac{\partial}{\partial x_i}\right)^k f(t\boldsymbol{x}).$$

断言之证明: 对 $k \in \mathbb{Z}^+$ 使用数学归纳法,

当 $k = 1$ 时, 由于 $t\boldsymbol{x} = (tx_1, tx_2 \cdots, tx_n)$, 故

$$\frac{\partial f(t\boldsymbol{x})}{\partial t} = \sum_{i=1}^{n} \frac{\partial f(t\boldsymbol{x})}{\partial x_i} \cdot \frac{\partial (tx_i)}{\partial t} = \left(\sum_{i=1}^{n} x_i \frac{\partial}{\partial x_i}\right) f(t\boldsymbol{x}).$$

即命题对 $k = 1$ 成立. 假设命题对 $k \geqslant 1$ 成立, 对 $k+1$: 由归纳假设,

$$\frac{\partial^k f(t\boldsymbol{x})}{\partial t^k} = \left(\sum_{i=1}^{n} x_i \frac{\partial}{\partial x_i}\right)^k f(t\boldsymbol{x}),$$

故

$$\begin{aligned}
\frac{\partial^{k+1} f(t\boldsymbol{x})}{\partial t^{k+1}} &= \frac{\partial}{\partial t}\left(\frac{\partial^k f(t\boldsymbol{x})}{\partial t^k}\right) = \frac{\partial}{\partial t}\left(\left(\sum_{i=1}^{n} x_i \frac{\partial}{\partial x_i}\right)^k f(t\boldsymbol{x})\right) \\
&= \sum_{i=1}^{n} \frac{\partial}{\partial x_i}\left(\left(\sum_{i=1}^{n} x_i \frac{\partial}{\partial x_i}\right)^k f(t\boldsymbol{x})\right) \cdot \frac{\partial (tx_i)}{\partial t} \\
&= \sum_{i=1}^{n} x_i \frac{\partial}{\partial x_i}\left(\left(\sum_{i=1}^{n} x_i \frac{\partial}{\partial x_i}\right)^k f(t\boldsymbol{x})\right) \\
&= \left(\sum_{i=1}^{n} x_i \frac{\partial}{\partial x_i}\right)^{k+1} f(t\boldsymbol{x}),
\end{aligned}$$

从而命题对 $k+1$ 也成立, 由数学归纳法知**断言**得证.

将等式 $f(t\boldsymbol{x}) = t^K f(\boldsymbol{x})$ 两边对 t 求 k 阶导, 由**断言**知

$$K(K-1)\cdots(K-k+1) t^{K-k} f(\boldsymbol{x}) = \frac{\partial^k f(t\boldsymbol{x})}{\partial t^k} = \left(\sum_{i=1}^{n} x_i \frac{\partial}{\partial x_i}\right)^k f(t\boldsymbol{x}),$$

再令 $t=1$ 得

$$K(K-1)\cdots(K-k+1)f(\boldsymbol{x})=\left(\sum_{i=1}^{n}x_i\frac{\partial}{\partial x_i}\right)^k f(\boldsymbol{x}).$$

14.26. 设函数 $z=\mathrm{e}^{xy^2}$, 其中 $x=t\cos t, y=t\sin t$, 求 $\left.\dfrac{\mathrm{d}z}{\mathrm{d}t}\right|_{t=\frac{\pi}{2}}$.

解答:

$$\frac{\mathrm{d}z}{\mathrm{d}t}=\frac{\partial z}{\partial x}\cdot\frac{\mathrm{d}x}{\mathrm{d}t}+\frac{\partial z}{\partial y}\cdot\frac{\mathrm{d}y}{\mathrm{d}t}=y^2\mathrm{e}^{xy^2}\cdot(\cos t-t\sin t)+2xy\mathrm{e}^{xy^2}\cdot(\sin t+t\cos t),$$

当 $t=\dfrac{\pi}{2}$ 时, $x=0, y=\dfrac{\pi}{2}$, 从而 $\left.\dfrac{\mathrm{d}z}{\mathrm{d}t}\right|_{t=\frac{\pi}{2}}=-\dfrac{\pi^3}{8}$.

14.27*. 设函数 $u=z\sin\dfrac{y}{x}$, 其中 $x=3r^2+2s, y=4r-2s^3, z=2r^2-3s^2$, 求 $\dfrac{\partial u}{\partial r}$ 和 $\dfrac{\partial u}{\partial s}$.

解答: 直接计算即可, 计算结果略.

14.28. 设函数 $x=r\cos\alpha-t\sin\alpha, y=r\sin\alpha+t\cos\alpha$, 其中 $\alpha\in\mathbb{R}$ 为常数, 求证: 对任意可微函数 $f(x,y)$, 都有

$$\left(\frac{\partial f}{\partial x}\right)^2+\left(\frac{\partial f}{\partial y}\right)^2=\left(\frac{\partial f}{\partial r}\right)^2+\left(\frac{\partial f}{\partial t}\right)^2.$$

解答: 由于

$$\frac{\partial f}{\partial r}=\frac{\partial f}{\partial x}\cdot\frac{\partial x}{\partial r}+\frac{\partial f}{\partial y}\cdot\frac{\partial y}{\partial r}=\frac{\partial f}{\partial x}\cdot\cos\alpha+\frac{\partial f}{\partial y}\cdot\sin\alpha,\quad \frac{\partial f}{\partial t}=-\frac{\partial f}{\partial x}\cdot\sin\alpha+\frac{\partial f}{\partial y}\cdot\cos\alpha,$$

故

$$\left(\frac{\partial f}{\partial r}\right)^2+\left(\frac{\partial f}{\partial t}\right)^2=\left(\frac{\partial f}{\partial x}\cdot\cos\alpha+\frac{\partial f}{\partial y}\cdot\sin\alpha\right)^2+\left(-\frac{\partial f}{\partial x}\cdot\sin\alpha+\frac{\partial f}{\partial y}\cdot\cos\alpha\right)^2=\left(\frac{\partial f}{\partial x}\right)^2+\left(\frac{\partial f}{\partial y}\right)^2.$$

14.29*. 设函数

$$\boldsymbol{f}(x,y)=\begin{pmatrix}u(x,y)\\ v(x,y)\end{pmatrix}=\begin{pmatrix}\dfrac{x}{x^2+y^2}\\ \dfrac{-y}{x^2+y^2}\end{pmatrix},$$

求证: $\boldsymbol{f}(x,y)$ 是 $\mathbb{R}^2\backslash\{(0,0)\}$ 到自身的 C^1 同胚映射, 并求 $\boldsymbol{f}(x,y)$ 在 $(x_0,y_0)\in\mathbb{R}^2\backslash\{(0,0)\}$ 处的雅可比行列式.

解答: 由于 $\boldsymbol{f}(x,y)$ 将 $(r\cos\theta, r\sin\theta)$ 映成 $\left(\dfrac{1}{r}\cos\theta, -\dfrac{1}{r}\sin\theta\right)$, 即在极坐标系下将 (r,θ) 映成 $\left(\dfrac{1}{r}, -\theta\right)$, 其中 $r\in\mathbb{R}^+, \theta\in\mathbb{R}$, 故 $\boldsymbol{f}(x,y)$ 是 $\mathbb{R}^2\backslash\{(0,0)\}$ 到自身的 C^1 同胚映射 , 且

$$\det\left(\boldsymbol{J}_{\boldsymbol{f}}(x_0,y_0)\right)=\begin{vmatrix}-\dfrac{x_0^2-y_0^2}{(x_0^2+y_0^2)^2} & -\dfrac{2x_0y_0}{(x_0^2+y_0^2)^2}\\ \dfrac{2x_0y_0}{(x_0^2+y_0^2)^2} & -\dfrac{x_0^2-y_0^2}{(x_0^2+y_0^2)^2}\end{vmatrix}=\frac{1}{(x_0^2+y_0^2)^2}.$$

14.30. 求下列函数的高阶偏导数 $\dfrac{\partial^{\sum\limits_{i=1}^{n}m_i}f(\boldsymbol{x})}{\partial x_1^{m_1}\partial x_2^{m_2}\cdots\partial x_n^{m_n}}$, 其中 $m_i\geqslant 0, \forall 1\leqslant i\leqslant n$:

(1) $f(x_1,x_2,\cdots,x_n)=\mathrm{e}^{\sum\limits_{i=1}^{n}x_i}$;

(2) $f(x_1,x_2,\cdots,x_n)=\ln\left(\sum\limits_{i=1}^{n}a_ix_i\right)$, 其中 $a_i\,(1\leqslant i\leqslant n)$ 为常数.

解答:

(1) 直接计算得

$$\frac{\partial^{\sum\limits_{i=1}^{n} m_i} f(\boldsymbol{x})}{\partial x_1^{m_1}\partial x_2^{m_2}\cdots\partial x_n^{m_n}} = \mathrm{e}^{\sum\limits_{i=1}^{n} x_i}.$$

(2) 直接计算得

$$\frac{\partial^{\sum\limits_{i=1}^{n} m_i} f(\boldsymbol{x})}{\partial x_1^{m_1}\partial x_2^{m_2}\cdots\partial x_n^{m_n}} = \frac{(-1)^{\sum\limits_{i=1}^{n} m_i-1}\prod\limits_{i=1}^{n} a_i^{m_i}\left(\sum\limits_{i=1}^{n} m_i-1\right)!}{\left(\sum\limits_{i=1}^{n} a_i x_i\right)^{\sum\limits_{i=1}^{n} m_i}}.$$

14.31*. 设函数 f 具有二阶连续偏导数, 求下列函数的二阶偏导数:

(1) $z=f(x^2+y^2,xy)$;

(2) $u=f(x_1+x_2+\cdots+x_n)$.

解答:

(1) 直接计算得

$$\frac{\partial^2 z}{\partial x^2} = 4x^2 f''_{11} + 4xy f''_{12} + y^2 f''_{22} + 2f'_1;$$

$$\frac{\partial^2 z}{\partial x\partial y} = 4xy f''_{11} + 2(x^2+y^2) f''_{12} + xy f''_{22} + f'_2;$$

$$\frac{\partial^2 z}{\partial y^2} = 4y^2 f''_{11} + 4xy f''_{12} + x^2 f''_{22} + 2f'_1.$$

(2) 直接计算得

$$\frac{\partial^2 u}{\partial x_i\partial x_j} = f''(x_1+x_2+\cdots+x_n), \forall\, 1\leqslant i,j\leqslant n.$$

14.32. 验证下列函数满足 Laplace 方程 $\dfrac{\partial^2 z}{\partial x^2}+\dfrac{\partial^2 z}{\partial y^2}=0$:

(1) $z=\arctan\dfrac{y}{x}$;

(2) $z=\ln\sqrt{x^2+y^2}$.

解答:

(1) 直接计算得

$$\frac{\partial^2 z}{\partial x^2} = \frac{2xy}{(x^2+y^2)^2}, \quad \frac{\partial^2 z}{\partial y^2} = -\frac{2xy}{(x^2+y^2)^2},$$

故 $\dfrac{\partial^2 z}{\partial x^2}+\dfrac{\partial^2 z}{\partial y^2}=0$.

(2) 直接计算得

$$\frac{\partial^2 z}{\partial x^2} = \frac{y^2-x^2}{(x^2+y^2)^2}, \quad \frac{\partial^2 z}{\partial y^2} = \frac{x^2-y^2}{(x^2+y^2)^2},$$

故 $\dfrac{\partial^2 z}{\partial x^2}+\dfrac{\partial^2 z}{\partial y^2}=0$.

14.33*. 验证函数 $u=(x_1^2+x_2^2+x_3^2+x_4^2)^{-1}$ 满足 Laplace 方程 $\displaystyle\sum_{i=1}^{4}\frac{\partial^2 u}{\partial x_i^2}=0$.

解答: 直接计算得

$$\frac{\partial^2 u}{\partial x_i^2} = -2\cdot\frac{x_1^2+x_2^2+x_3^2+x_4^2-4x_i^2}{(x_1^2+x_2^2+x_3^2+x_4^2)^3}, \forall\, 1\leqslant i\leqslant 4,$$

故 $\displaystyle\sum_{i=1}^{4}\frac{\partial^2 u}{\partial x_i^2}=0$.

14.34. 设 $f(x)$ 二阶可导, 求证: $F(x,t)=\dfrac{1}{2}(f(x-ct)+f(x+ct))$ (c 为常数) 满足方程: $\dfrac{\partial^2 F}{\partial t^2}=c^2\dfrac{\partial^2 F}{\partial x^2}$.
解答: 直接计算得

$$\frac{\partial^2 F}{\partial t^2}=\frac{c^2}{2}(f''(x-ct)+f(x+ct)),\quad \frac{\partial^2 F}{\partial x^2}=\frac{1}{2}(f''(x-ct)+f(x+ct)),$$

故 $\dfrac{\partial^2 F}{\partial t^2}=c^2\dfrac{\partial^2 F}{\partial x^2}$.

14.35*. 求证: 在极坐标变换 $\begin{cases} x=r\cos\theta \\ y=r\sin\theta \end{cases}$ 下, Laplace 方程 $\dfrac{\partial^2 u}{\partial x^2}+\dfrac{\partial^2 u}{\partial y^2}=0$ 的形式为

$$\frac{\partial^2 u}{\partial r^2}+\frac{1}{r}\cdot\frac{\partial u}{\partial r}+\frac{1}{r^2}\cdot\frac{\partial^2 u}{\partial \theta^2}=0.$$

解答: 直接计算得

$$\frac{\partial u}{\partial r}=\frac{\partial u}{\partial x}\cdot\cos\theta+\frac{\partial u}{\partial y}\cdot\sin\theta,\quad \frac{\partial^2 u}{\partial r^2}=\frac{\partial^2 u}{\partial x^2}\cdot\cos^2\theta+\left(\frac{\partial^2 u}{\partial y\partial x}+\frac{\partial^2 u}{\partial x\partial y}\right)\cdot\sin\theta\cos\theta+\frac{\partial^2 u}{\partial y^2}\cdot\sin^2\theta,$$

$$\frac{\partial^2 u}{\partial \theta^2}=\frac{\partial^2 u}{\partial x^2}\cdot r^2\sin^2\theta-\left(\frac{\partial^2 u}{\partial x\partial y}+\frac{\partial^2 u}{\partial y\partial x}\right)\cdot r^2\sin\theta\cos\theta+\frac{\partial^2 u}{\partial y^2}\cdot r^2\cos^2\theta-\frac{\partial u}{\partial x}\cdot r\cos\theta-\frac{\partial u}{\partial y}\cdot r\sin\theta,$$

故

$$\frac{\partial^2 u}{\partial r^2}+\frac{1}{r}\cdot\frac{\partial u}{\partial r}+\frac{1}{r^2}\cdot\frac{\partial^2 u}{\partial \theta^2}=\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=0.$$

14.36. 设函数 $u(x,y,z)=\dfrac{x-y+z}{x+y-z}$, 求证:
(1)

$$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}+z\frac{\partial u}{\partial z}=0;$$

(2)

$$x^2\frac{\partial^2 u}{\partial x^2}+y^2\frac{\partial^2 u}{\partial y^2}+z^2\frac{\partial^2 u}{\partial z^2}+2xy\frac{\partial^2 u}{\partial x\partial y}+2xz\frac{\partial^2 u}{\partial x\partial z}+2yz\frac{\partial^2 u}{\partial y\partial z}=0.$$

解答:
(1) 在**14.25**的结论中取 $n=3,K=0,k=1$ 即可.
(2) 在**14.25**的结论中取 $n=3,K=0,k=2$ 即可.

14.37*. 设 $x=2r-s,y=r+2s$, 且函数 $f(x,y)$ 具有二阶连续偏导数, 求 $\dfrac{\partial^2 f(x,y)}{\partial r\partial s}$.
解答: 直接计算得

$$\frac{\partial^2 f(x,y)}{\partial r\partial s}=-2f''_{xx}+3f''_{xy}+2f''_{yy}.$$